

Apollo Care is a Chicago-based company using data analysis to support pharmaceutical companies. The company asked the University of Chicago Data Science Clinic to develop an algorithm to estimate patient insurance deductibles and reset dates using three years of claims data. Identifying these moments allows companies to determine when patients face the highest out-of-pocket costs and offer coupons or copay assistance.

As many patients do not use their insurance consistently enough to produce a predictable deductible pattern, the team first filtered out individuals with insufficient claim activity or non-commercial insurance plans. The team then used key elements of each individual members' approaches in finding the **inflection point**; the claim at which a patient's out-of-pocket payment drops sharply because their deductible has been met, to create one combined model.

This model was then able to estimate deductibles and reset dates for **58.7% of patients**. These patients exhibited clear deductible and reset date patterns, similar to the example shown in the figure below, where the percentage of expenses paid by the patient fell sharply once the deductible is hit. Estimating these amounts and reset dates helps Apollo Care identify when patients are likely to face the highest out-of-pocket costs and connect them with effective financial assistance programs.

